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A reasoning based explainable multimodal fake news detection for low resource language using large language models and transformers

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Abstract

Nowadays, individuals rely predominantly on online social media platforms, news feeds, websites, and news aggregator applications to acquire recent news stories. This trend has resulted in an increase in the number of available social media platforms, online news feeds, and news aggregator applications. These news platforms have been accused of spreading fake news to gain more attention and recognition. Earlier, this misinformation or fake news used to be propagated only in the text form. However, with the advent of technology, now it is spread in multimodal forms, such as images with text, videos, and audio with textual content. Currently, the automatic fake news detection models are focused on high resource languages and superficial output. Social media users need clarity and reasoning when it comes to identifying fake news, rather than just a superficial classification of news as fake. Providing context, reasoning, and explanations can help users understand why certain news is misleading or false. Hence, a multimodal system has to be developed to identify and justify fake news. In this proposed work, we have developed a multimodal fake news system for the Low Resource Language Tamil with reasoning-based explainability. The dataset for this proposed work is retrieved from fact-check websites and official news websites. We have experimented with different combinations of models for visual and text modalities. Further, we integrated LLM-based image descriptions into our model with the text and visual features, resulting in an F1 score of 0.8736. We used the Siamese model to determine the similarity of the news and its image descriptions. Additionally, we conducted error analysis and used explainable artificial intelligence to explore the reasoning behind our model's predictions. We also present the textual reasoning for the model's predictions and match them with images.

Keywords: Misinformation, Multimodal fake news, Tamil, Transformers, mBERT, XLMRoBERTa, ViT, DeiT, XAI

Introduction

The term 'multimodal misinformation' refers to disseminating false information through various communication channels, such as text and image, text and audio, and text and video, to create compelling yet misleading content [1]. In this work, we will use the term 'fake news' to describe misinformation, the most commonly used phrase. Technological advancements have enabled fake news to become more realistic and deceptive. With advanced technologies and platforms, creating and distributing misleading content has become more accessible, leading to an increase in multimodal fake news. The ease of creating and spreading multimodal fake news is due to advancements in image creation, video manipulation, and deepfake technologies, empowering individuals with malicious intent.

The textual content provides essential information, offering context to support the visuals and enhance the credibility of multimodal fake news [2]. It is possible to create manipulated textual information to resemble legitimate news reports that imitate the style and structure of reputable sources [3]. News consumers are often misled by convincing information that combines manipulated text with false images, audio, or video, increasing the impact of fake news. Similarly, images play a crucial role in multimodal fake news by providing visual proof for propagating false information [4]. Fabricated or false images can be created using advanced editing tools and generative AI techniques. For example, an image can be manipulated to include people who were not present at a specific event or to depict events that never occurred. Manipulated images are often shared on social media, rapidly spreading and influencing public opinion [5]. Moreover, according to the literature studies, fake information spreads faster than reliable information [6], and texts with visual content spread faster and get more likes and shares [7].

The rise of DeepFake technology has increased the danger of multimodal fake news. DeepFake involves creating synthetic images/videos using an AI algorithm that overlays the face and voice of one person onto another, giving the impression that the person in the video is performing actions they never actually did in real life. These DeepFakes can produce falsified speeches, interviews, or even footage of public figures that are challenging to identify through standard analysis.

Similarly, advanced audio editing tools can modify or fabricate recordings, creating false narratives or misrepresenting individuals' words. The deception becomes even more sophisticated by combining manipulated audio, visuals, or text. The consequences of multimodal fake news are far-reaching, as it can be used for political propaganda, spreading disinformation, and influencing public opinion. Spreading fake information quickly through social media can strengthen conspiracy theories, decrease trust in reliable news, and even lead to violence or unrest. So, fighting fake news requires various approaches, which means creating better detection technologies to find fake media in different forms. It is essential to have media literacy programs to educate the public about the prevalence and risks of various forms of fake news and to teach people how to evaluate the credibility of information they come across critically. Social media platforms and content-sharing websites should enforce strict policies and algorithms to identify and remove deceptive and manipulated content.

Moreover, regarding a particular language like Tamil, we don't have datasets comprising multimodal fake news. To develop a multimodal system, we should have a dataset that can be used to identify multimodal fake news in Tamil. We have collected the Tamil Multimodal Fake News dataset from fact-checked websites (https://youturn.in) and the official news website (https://www.dinamalar.com/).

All these indications imply the necessity for an automated system to verify the credibility [8] of information circulated or uploaded on social media platforms. In this proposed work, we study multimodal fake news detection on our proposed dataset, which consists of samples collected from fact-check and official news websites. The dataset consists of news with Headlines, Information, and Images and is labeled into 3-way classes namely 'FAKE', 'FALSE', and 'TRUE'. The difference between the fake and false classes is that the former is completely fake because of the fabricated image or text in the content, whereas the latter has content that has some information that is true but is connected with some other information to make it look legible. Moreover, In this proposed work, we use explainable artificial intelligence to find the reason behind the model's prediction.

Major contributions in this proposed work

In this proposed work, we have collected a Multimodal Fake News Dataset for the Tamil language and developed a system for detecting the same. We have used the transformer-based model and the Large Language Model. The following are the major contributions to this work.

- We have developed a Low Resource Language Multimodal Fake News Dataset for Tamil, which consists of data collected from fact-check and news websites.
- We have developed a baseline system for the Multimodal Fake News Detection using the transformer-based model.
- We have used a Siamese model in determining the similarity of news and its image in understanding fake news.
- We used the LLMs to generate the description for the images of supporting articles for fake news, which is added to train the model along with the multimodal news.
- We have experimentally evaluated the different proposed models and their performance for the dataset and conducted the error analysis.
- We used the Explainable Artificial Intelligence (XAI) method with reasoning to evaluate the proposed model and provide image-based explanations.

The remainder of the paper is organized as follows. First, in "Related works" section, we briefly review existing works on multimodal and unimodal systems that work with the same and different datasets and methodologies. Followed by "Dataset description and evaluation metrics" section, which explains the dataset and the evaluation metrics we have used. Then, in "Proposed method" section, we describe feature extraction and the proposed methodologies of LLM generated descriptions and siamese networks for the multimodal fusion model, followed by the experiments, results, and discussions in "Experiment results and discussion" section. Then, in "Error analysis of proposed models and XAI-based reasoning" section, we study the error analysis and discuss the XAI based reasoning on the proposed models. Finally, we conclude with hints on possible future works in "Conclusions and future works" section.

Related works

Social media has become the primary source of information for many people, leading to a significant increase in the volume of information shared across various social media platforms. However, a significant concern is the authenticity of the information being disseminated. Existing literature discusses traditional machine learning techniques to detect fake news in text [9] and multimodal data. Despite this, there remains a need for further improvement, especially as the issue of multimodal fake information continues to grow. Our extensive literature review covers different methods for detecting fake news using textual and multimodal data, the various methodologies employed for identifying multimodal fake news, and the works on Dravidian languages like Tamil and Malayalam. The following section provides more detail about these methods and the data that they have used.

Textual fake news detection

The work by Alkaddour et al. [10] has used the sentiments of news image captions and the image-text features obtained using a multilayer perception. They have used multiple logistic regression to show that the considered sentiment features are significant in predicting the outcome. Cui et al. [11] have used the user's comments from the posts and their latent sentiments in identifying fake news. They have incorporated latent sentiments into an end-end deep learning framework which they named SAME. Boididou et al. [12] have presented a comparative study on fake news detection which is spreading through the popular social media network Twitter. They have identified methods like textual patterns, same-topic tweets, and a semi-supervised learning scheme that leverages the decisions of two independent classifiers. Mehta et al. [13] has proposed a fake news detection model on the transformers-based BERT model, which they fine-tuned for specific domain datasets and used human justification and metadata.

Multimodal fake news detection

In their work, Uppada et al. [14] has developed a multimodal framework comprising different architectures to extract visual and linguistic features from the data. They utilized a variety of image-related models, including VGG19, ResNET50, Inception, and Xception, to assess the polarity and likelihood of image manipulation within the dataset. The researchers employed models such as LSTM, CNN, BiGRU, CapsuleNet, 2D CNN, BERT, and RoBERTa for image captions. Subsequently, the most effective models from the experiments above were combined to create a final model for multimodal fake news detection. The work by Li et al. [15] has introduced an Entity-oriented Multimodal Alignment and Fusion Network (EMAF) for detecting fake news in multimodal news. Unlike existing methods that operate at a fine-grained or coarse-grained level, EMAF performs cross-modal interaction at the entity level. The model consists of an Encapsulating module, a Cross-modal Alignment module, a Cross-modal Fusion module, and a final Classifier. The Alignment module aligns visual and textual entities using an improved dynamic routing algorithm, while the Fusion module captures the consistencies and differences among aligned entities and textual entities. The Classifier makes the final prediction based on the fused representation. Experimental results demonstrate that EMAF outperforms previous methods in detecting fake news.

Wu [16] has introduced a method called Human Cognition-based Consistency Inference Networks (HCCIN), inspired by the cognitive process. They have applied HCCIN to thoroughly investigate the consistent and inconsistent detection of multimodal fake news semantics. They developed a cross-modal alignment layer to understand the semantics between textual and visual information. Additionally, they included a layer for comment clues to help identify the most relevant semantics.

Guo [17] proposed a mutual attention-based fusion network for multimodal fake news detection. They developed a mutual attention neural network (MANN) to learn the relationships between different modalities. The network primarily consists of four components to identify multimodal fake content: feature extractor, mutual attention fusion, fake news detector, and event discriminator.

The work by Wang et al. [18] introduces a method for detecting fake news using two-level visual features and multimodal transformer models. They utilized features from image captions and images, treating them as sequences for processing the transformer models. The two-level visual features, comprising global and entity-level features, were employed to enhance the model's performance. The multimodal transformer facilitated interaction between the different modalities, and the output was passed to fully connected layers for the final prediction.

The work by Zhou et al. [19] has developed a multi-grained information fusion model for detecting multimodal fake news. They utilized a transformer-based model to encode token-level features from both text and images. Additionally, they combined these features with coarse-grained features encoded by the CLIP encoder. To address the ambiguity problem, they used unimodal networks with similarity-based weighting. Jing et al. [20] have proposed a multimodal progressive fusion network that captures the information of each modality at the same level and different levels using a mixer to establish a strong connection between the modalities. Wang et al. [21] proposed a model for detecting multimodal cyberbullying posts on social networks. They utilized textual features and hierarchical attention networks to extract session features in social networks and different multimedia.

Hua et al. [22] proposed a BERT-based model incorporating back translation and an entire image multimodal model using contrastive learning. They used back-translated features, including textual and visual features, and contrastive learning to derive more reasonable multimodal representations. Khattar et al. [23] have proposed a multimodal framework that uses auto-encoder coupled with a binary classifier. They have performed the evaluation of the model on Weibo and Twitter data. The work by [24] has introduced a bidirectional cross-modal attention model that combines textual and image features bidirectionally. The model consists of three modules: image2text fusion, text2image fusion, and prediction. They evaluated the model using datasets from Weibo, Twitter, Politi, and Gossip.

Aneja et al. [25] have proposed a method that detects out-of-context images and text-pairs from social media posts. They check for the captions to see whether they correspond to the same objects in the image but are semantically different. They evaluated the performance using the dataset created from various news websites, blogs, and social media posts. [26] have proposed a method for automatically generating and annotating synthetic multimodal misinformation due to the data-intensive nature of deep neural networks and the difficulty in human annotation. [27] have explored contrastive learning in the domain of fake news detection and have developed a self-learning model, and experimented on publicly available datasets.

Multilingual fake news detection

The work by [28] has studied the impact of transformers on identifying fake news in Malayalam and Tamil news data. They have created this dataset which is translated from the data retrieved from FakeNewsAMT and Celebrity Fake News covering various domains like sports, politics, and science. They have extended the work by [29] where they have developed the dataset from the existing one and offered a multilingual dataset for five low-resource languages. The work by [30] is on multilingual datasets that are discourse aware, whereas the work by [31] is addressing the internationality and multiliguality in fake news detection. Furthermore, we can see that there is not much multilingual work is done for the low-resource language Tamil.

The above studies on recent works suggest that there is a lack of a system that could correctly identify multimodal fake news, and there is room for improvement from the existing system especially in the case of low-resource languages.

Dataset description and evaluation metrics

In this proposed work, we created a Multimodal Fake News dataset for Tamil Language. We have used Python-based web crawling to collect data from fact-checking and official news websites to retrieve the same.

We have collected data from a fact-checking website called Youturn (https://youturn.in) and the official news website called Dinamalar (https://www.dinamalar.com/) for Tamil-verified news. The data collected is from 2022 to 2024 till April. As the number of fake news officially verified is less than the actual news, we have taken an equal number of samples for the dataset creation. Moreover, when we collected the verified fact-checked data, we had two samples called 'FAKE' and 'FALSE'. The class 'FAKE' corresponds to data where the news and the images are completely manipulated and hence they are completely fake, whereas the 'FALSE' corresponds to the class where the news or the image may be real but it is combined with a piece of different news to make it look real. The number of samples in the dataset is given in Table 1 below with different classes, along with the sample from each class in Fig. 1.

Class 'FALSE' labeling is based on the textual news and the corresponding image, where the news may not be completely fake. For example, in the given Fig. 1, the class 'FALSE' briefs about the news of an actress who is said to be the daughter of an actor named 'Nambiar'. Actually, she is not the daughter of the so-called actor, but her father's name is also the same, belonging to the community 'Nambiar.' Here, we can say that the news is 'FAKE' from a broader perspective, which is not entirely fake, because here 'Nambiar' is also the name of the actor's father, but who is not the film actor mentioned in the news. We could have gone for a simple binary classification, considering this also as a 'FAKE' class. However, when it comes to social media use, we can see this type of post is commonly encountered and is generally tagged as fake or misinformation. Thus, if we analyze them, we can say that there might be a part of the news that is true which is combined with another to make it look real. We have added more samples from this 'FALSE' class in the Fig. 2 along with their news and image. Similar to the above sample discussed above, the labeling is done based on the information from the data. We have collected data from fact-checking and official news websites, allowing us to access

Headline: தமிழகத்தில் ஒரே கட்டமாக ஏப்.,19ல் ஓட்டுப்பதிவு

News: புதுடில்லி: தமிழகத்தில் ஒரே கட்டமாக ஏப்.,19ம் தேதி லோக்சபா தேர்தல் நடத்தப்படும் என தேர்தல் ஆணையம் அறிவித்து உள்ளது. இடைத்தேர்தல்விளவங்கோடு தொகுதிக்கு இடைத்தேர்தல்- ஏப்,19ம் தேதி…

Class: ***TRUE***

Headline: நாம் தமிழர் கட்சியின் 234 வேட்பாளர்களும் எந்த குற்ற பின்னணியும் இல்லாதவர்கள் – தேர்தல் ஆணையம்.

News: 2021ம் ஆண்டு சட்டசபை தேர்தலிலும் நாம் தமிழர் கட்சி தனித்து களம் காண்கிறது. அக்கட்சியின் ஒருங்கிணைப்பாளர் 234 வேட்பாளர்களையும் ஒரே மேடையை அறிமுகம் செய்து இருந்தார்….அதுவும் தவறான தகவலேனக் கூறப்பட்

Class: ***FAKE***

Headline: விஜய் படம் முதல் சன் டிவி சீரியல் வரை நடித்த இவர் தான் பிரபல நடிகர் நம்பியாரின் மகள்! ..வேடத்தில்நடித்திருக்கிறார்.

News: பிரபல நடிகர் நம்பியாரின் மகள் தான் சீரியல் நடிகை சினேகா என்று கூறி புகைப்படம் ஒன்று சமூக ஊடகங்களில் வைரலாகப் பரவி வருகிறது…, செல்லமே போன்ற சீரியல்களில் நடித்திருக்கிறார்" என்று குறிப்பிட்டும் சமூக ஊடகங்களில் பரவி வருவதைக் காண முடிகிறது

Class: FALSE

Fig. 1 Dataset samples with headline, news, images and class labels

Table 1 Classwise dataset details

Data	#Samples	#TRUE	#FAKE	#FALSE
Train	2625	1312	1042	271
Validation	525	263	208	54
Test	350	175	139	36
Total	3500	1750	1389	361

real-world fact-checked news articles. Consequently, our dataset is closely aligned with actual events. Additionally, we ensured that the authentic news articles were collected from the same period as the fact-checked articles, which will help ensure consistency in the information. Moreover, we have considered the three-way classification of the samples, which will help categorize the news better than the traditional binary classification.

Evaluation metrics

This proposed work focuses on accurately classifying news posts/articles into different classes. Accordingly, we use the following evaluation measures.

1. *Precision*: is the proportion of correct classification to the number of incorrect classification.

Headline: முதல்வர் ஸ்டாலின் அவர்களே சொன்னா கோச்சிக்கிறீங்க இருந்தாலும் சொல்லித்தான் ஆகணும். உங்க அரசாங்கம் உண்மையிலேயே கையாளாகாத அரசாங்கம் தான்.

News: இந்த பள்ளி மாணவிகளுக்கு எந்த பள்ளியில்\\ படிக்கின்றோம் என்பதே தெரியவில்லை, ...

Class: FALSE

Headline: மழையின் கோர பிடியில் நெல்லை.

News: குமரிக்கடல் மற்றும் அதனை \\ஒட்டியுள்ள பகுதிகளில் வளிமண்டல கீழ் அடுக்கு சுழற்சி நிலவுவதன் காரணமாகத் தூத்துக்குடி, திருநெல்வேலி, கன்னியாகுமரி தென்காசி ஆகிய மாவட்டங்களில்...

Class: FALSE

Headline: இன்று மேட்ச்… My Man எனக் குறிப்பிட்டு சுப்மன் கில்லுக்கு சாரா டெண்டுல்கர் வாழ்த்து..

News: இந்தியாவில் நடைபெற்று வரும் 13-வது உலகக்கோப்பை கிரிக்கெட் தொடரில், இந்தியா மற்றும் நியூசிலாந்து அணிகளுக்கு இடையிலான முதல் அரையிறுதி போட்டி இன்று மும்பை வான்கிடே மைதானத்தில் நடைபெறவுள்ளது....

Class: FALSE

Fig. 2 Additional samples on 'FALSE' class

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

2. *Recall*: is the proportion of correct classification to the number of missed entries.

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

3. *F1-Score*: is the harmonic mean of precision and recall.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

Where True Positive (TP) corresponds to the number of samples correctly classified to the 'True' Class, True Negative (TN) accounts for the number of samples correctly classified to other Classes, False Positive (FN) considers the number of 'True' Class wrongly classified to other Classes and False Negative (FN) reflects the number of different classes incorrectly assigned to 'True' class. We have calculated the macro-average for precision, recall, and F1.

Proposed method

In this proposed work, we have considered the fusion of multimodal features consisting of Images, Headlines, and News content.

Moreover, we have also experimented with the fusion of supporting images that justify how the news is fake, along with the multimodal features. Figure 3 shows the model architecture, where we have given a high-level view of our proposed system.

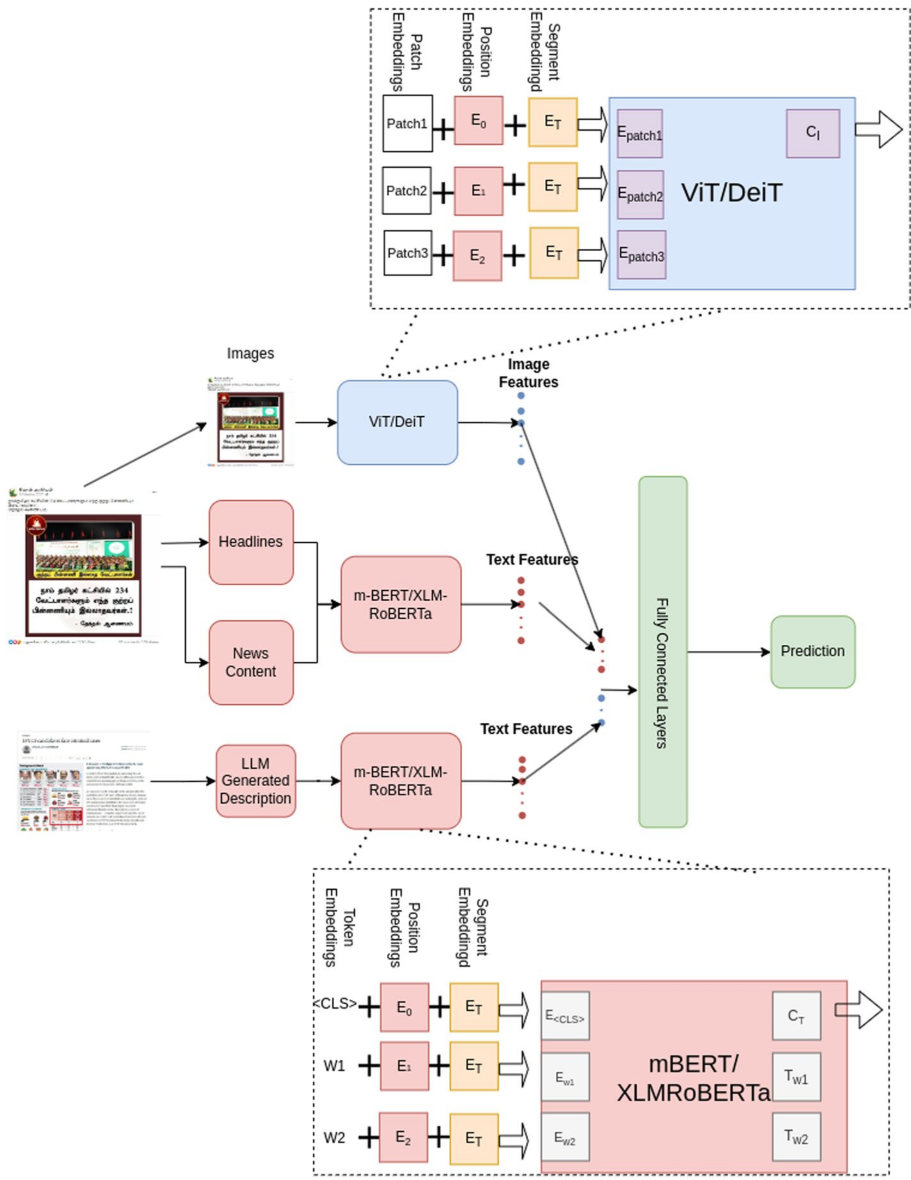


Fig. 3 Model architecture of the proposed method: images along with headlines, news, LLM descriptions are fed to the transformer-based models and the features from these are fed to fully connected layers to get the predictions. The expanded image in the top shows how the vision transformer is processed and below image shows how the text is processed by mBERT/XLMRoBERTa

Feature extraction

We need methods to extract features from multimodal data comprising images and text before using the fusion model. We use the transformer-based m-BERT multilingual version of BERT [32] to extract text features, and also we used the multilingual version of RoBERTa [33] (XLMRoBERTa). In BERT (Bidirectional Encoder Representations from Transformers), feature extraction refers to obtaining contextualized word representations from the model. BERT is a transformer-based model pre-trained on a large corpus of unlabeled text data. It learns to capture the contextual information of words by considering both the left and right contexts of each word in a sentence.

Text feature extraction

The feature extraction process in the proposed transformer-based model involves the following steps:

WordPiece Tokenization: m-BERT uses WordPiece tokenization, which involves breaking down words into smaller sub-word units. This helps in handling rare and unknown words. For example, the word 'unhappiness' might be broken down into 'un', '##happy', and '##ness'. The '##' indicates that the token is a continuation of a previous token.

Special tokens: m-BERT uses special tokens:

- CLS (classification token) is added at the beginning of the input text.
- SEP (separator token) is added at the end of each sentence or between sentences.

Input representation: BERT's input representation for each token is a combination of the following embeddings:

Token embeddings: The embeddings for each token are generated by the tokenizer.

Segment embeddings: These distinguish between different sentences in a pair (used in tasks like question-answering).

Position embeddings: These indicate the position of each token in the sentence, allowing BERT to understand the order of the tokens.

Word embeddings: Each token is assigned an initial token embedding, typically randomly initialized. During pre-training, BERT learns to refine these embeddings based on the surrounding context. These initial embeddings are combined with the segment and position embeddings.

Transformer layers: BERT uses the Transformer architecture, which is made up of multiple layers (12 layers for BERT Base and 24 layers for BERT Large). Each layer has:

Multi-Head Self-Attention: This mechanism allows each token to attend to every other token in the sentence, enabling BERT to understand context bidirectionally. Self-attention computes a weighted sum of values (V) using attention scores derived from the queries (Q) and keys (K).

Feed-Forward Neural Networks: After the self-attention mechanism, the output is passed through a feed-forward neural network, which consists of two fully connected layers with a ReLU activation in between.

Layer Normalization and Residual Connections: Each sub-layer (self-attention and feed-forward) is followed by layer normalization and a residual connection, which helps in stabilizing the training and improving the flow of gradients.

Output representations: Hidden States: For each token, BERT produces a sequence of hidden states (one for each layer). The hidden state from the final layer is typically used as the token representation, though intermediate layers can also be used for different tasks.

[CLS] Token: The hidden state corresponding to the [CLS] token in the final layer is often used as the aggregate sequence representation for classification tasks. This token is designed to capture the overall meaning of the sentence or input text.

Image feature extraction

In this proposed model, we utilize multimodal data consisting of both images and text. To prepare the image data for the model, we need to extract its features. Typically, convolutional neural networks (CNN) [34] are employed for image feature extraction, utilizing convolution layers. However, in this work, we use Vision Transformers (ViT) [35], a transformer-based pre-trained model. ViT employs a transformer-based architecture to capture and represent meaningful visual information from input images. This feature extraction method utilizes a self-attention mechanism to capture global dependencies and relationships among the image patches. The self-attention method enables feature extraction regardless of whether an object or thing is present in the image. Additionally, ViT aggregates global information from the outset and effectively propagates it to the upper layers of its architecture.

The feature extraction process in ViTs has the following steps:

Patch extraction: The initial step involves dividing the input image into smaller patches, each of which represents a local region of the image. These patches can be square or rectangular and are typically non-overlapping. The chosen patch resolution determines the size of each patch. For instance, if the patch resolution is 16x16 pixels, the image is divided into a grid of 16x16 patches. Usually, the patches are flattened into 1D sequences.

Divide the image into non-overlapping patches of size $P \times P$. For $I'_{norm} \in R^{H' \times W' \times C}$ and patch size $P \times P$, the number of patches N is

$$N = \frac{H' \times W'}{P \times P} \tag{4}$$

Each patch I_p is of size $P \times P \times C$. Also the patches are flattened into 1D vector, for each patch I_p, the flattened patch x_p is

$$x_p = flatten(I_p) \in R^{P \times P \times C} \tag{5}$$

Patch embeddings: Each patch is linearly projected to a lower-dimensional embedding space. This projection is typically implemented using a fully connected layer. The resulting patch embeddings and corresponding position embeddings are combined to form the input embeddings for the transformer encoder.

The flattened patch x_p is projected to a fixed dimensional embedding z_p as

$$z_p = W_p X_p + b_p \tag{6}$$

$W_p \in R^{D\times(P\times P\times C)}$ and $b_p \in R^D$, D is the embedding dimension.

Position embeddings: To obtain the position information, we add position embeddings to each patch. These embeddings encode the positional or spatial coordinates of each patch within the image, helping the model understand the relative locations of different patches. During training, position embeddings can be learned or predefined based on the patch size, and they are added to the patch embeddings obtained in the next step. The position embedding E is added to each patch embedding z_p as

$$z'_p = z_p + E_p \tag{7}$$

where $E_p \in R^D$ is the positional embedding for patch p.

Transformer encoder: The transformer encoder is a crucial component in ViTs that captures the interactions between different patches and learns meaningful representations. It consists of multiple layers, typically identical, each containing two sub-layers: self-attention and feed-forward neural networks. For each layer l in the transformer, the multi-head self-attention method computes

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{D_K}})V \tag{8}$$

where $Q = W_Q Z$, $K = W_K Z$, $V = W_V Z$, and W_Q, W_K and W_V are learned projection matrices. D_K is the dimension of key vectors.

Token pooling: After the transformer encoder processes all the layers, the resulting embeddings contain detailed information about the image. Token pooling is used to create a fixed-length representation for the entire image. Token pooling combines the embeddings of all the patches into a single representation. The most common approach is to calculate the mean or sum of the patch embeddings. This pooled representation provides a comprehensive understanding of the image as a whole. A learnable classification token $z_0 \in R^D$ is added as

$$Z' = [z_0; z'_1; z'_2; ...; z'_N] \tag{9}$$

Classification head: The pooled representation obtained from token pooling is fed into a classification head. The classification head is typically a fully connected layer or a series of layers followed by a softmax activation function. It maps the pooled representation to the desired number of output classes, enabling the model to make predictions on the given task, such as image classification.

$$y = softmax(W_c {z_o}^L + b_c) \tag{10}$$

which is the output corresponding to the [CLS] token after 'L' Layers (${z_o}^L$).

The feature extraction process in Vision Transformers (ViTs) does not rely on handcrafted feature extractors like CNNs. Instead, ViTs utilize the self-attention mechanism in the transformer, allowing the model to learn relevant features directly from the image patches. This enables ViTs to effectively model long-range dependencies and capture both local and global information in an image. The end-to-end feature extraction process in ViTs enables them to achieve state-of-the-art performance on various computer vision tasks. Several variants of ViTs have been proposed, such as DeiT [36, 37], and others, which introduce modifications to the architecture to improve performance, memory efficiency, or scalability.

Description of images using LLM

In this section, we will discuss the image description generated using LLM and how it is being used with our proposed model. We use the Gemini-Pro-Vision model by the Google Gemini [38] Team to generate the description for the supporting images of 'FAKE' and 'FALSE' class labels. The Gemini model is trained on multimodal data, and hence, it can accept different types of information as part of a prompt, including text, image, audio, and video. Gemini is excellent at analyzing images because it can respond to multimodal prompts. For example, you can ask the Gemini Pro Vision model to calculate the cost of individual items by combining multiple images, such as an image of a list of items with their prices. In this proposed work, we have used the capabilities of the Gemini vision model to generate the description for the images that support the news claim, which belongs to the 'FAKE' and 'FALSE' classes.

Gemini is built on top of the transformer-based architecture ViT and processes images, as discussed in the previous section. We have used the Gemini model for visual instruction tuning, where we will supply an image and a prompt. Thus, when we give an

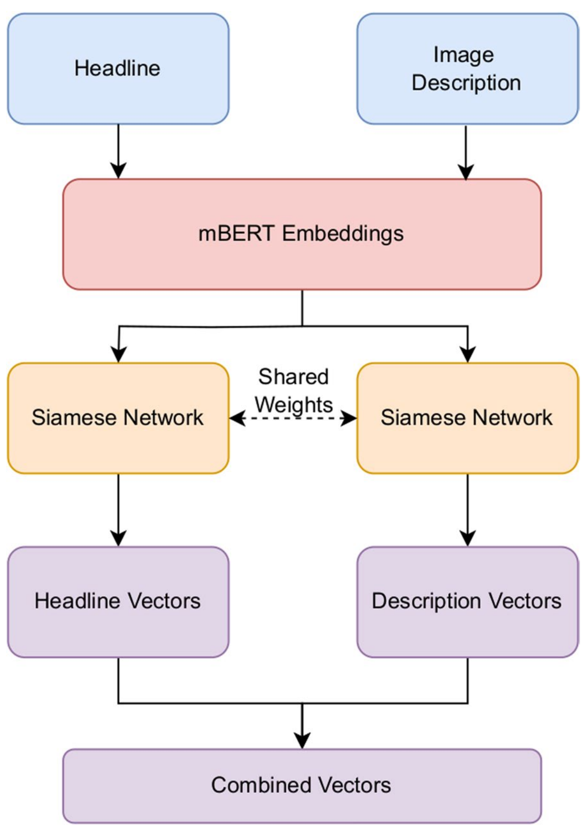


Fig. 4 Siamese network

image to the model, it will be divided into patches along with an encoded prompt, which is passed as an input. The model will generate a response from which the text content is retrieved and stored as a description for the image, summarized in Algorithm 1. A sample from LLM-generated descriptions and images (the original one from the dataset) is given in Example 1.

Example 1.

Description: *The image shows Indian Prime Minister Narendra Modi, along with two other men. One of the men has a pen in his shirt pocket., The original image shows a man in a white shirt with a pocket on the left side of his chest. The pocket is empty. The morphed image shows the same man with the same shirt, but the pocket is not empty.*

Algorithm 1 Algorithm for description of images using LLM

Input: Images I_m
Output: LLM Generated Text T_g

I_m=Open(img_path)
Initialize the LLM model L_m
Define Prompt P_t as "Describe what you see in the image."
response = $l_m.generate\_content([P_t, I_m])$
T_g = response.text

Feature extraction using Siamese network

A Siamese network is a specific type of neural network architecture that can used for processing text data. Siamese networks are commonly used in text similarity, paraphrase detection, and classification tasks. It comprises two or more identical subnetworks with the same parameters and weights. Each subnetwork processes input text data independently to learn and compare the representations. The output from each subnetwork is then combined to determine the similarity or dissimilarity between the input texts. Their ability to compare and process text representations makes them valuable for textual data tasks.

In this proposed work, we have incorporated text representations from the Siamese network as a feature for our model. We have used the headlines and LLM-generated descriptions of the images. As the descriptions have helped model understanding, we have taken them along with the headlines. As shown in Fig. 4, we get the mBERT embeddings from headlines and descriptions and pass them to the Siamese networks sharing the weights. The vectors or representations from the networks for headlines and descriptions are given as additional features of our proposed model.

Experiment results and discussion

This section explains the experiment results of the baseline model and proposed multimodal fusion models with headlines, news, and images. We have experimented with mBERT, XLMRoBERTa, another variant of BERT, and the DeiT variant of the vision transformer to see how they perform on multimodal fake news classification. As for the baseline model, we only use the fusion of news headlines, images, and news content. We have further experimented with the LLM-generated description of images that support the news also as a feature for the multimodal fusion model. The forthcoming section explains the baseline results of the proposed methods.

Baseline

We developed the baseline model using our proposed the multimodal fake news dataset from section "Dataset description and evaluation metrics". For the baseline system, we utilized a combination of news headlines, content, and images. The model was trained for ten epochs with callbacks on validation loss, using 1000 warmup steps, a batch size of 8, a learning rate of 3×10^{-5}, a maximum sequence length of 512 (aligned with mBERT's maximum sequence length), and ADAM optimizer [39] with a clip threshold of 1.0 and an epsilon value of 1×10^{-8}.

We will use two modalities, images and text, to classify fake news. Both modalities are represented as length 'L' vectors with dimensions d_I and d_T. In the case of transformer-based models, the length 'L' is limited to 512, so we maintain the length at 512. Smaller headlines or news will be padded with zeros to fit the size. We will consider the headline, content, and image as a modality pair. These paired representations will be simultaneously processed in one training step and undergo backpropagation simultaneously. The model's output will consist of class labels 0, 1, or 2, representing 'TRUE', 'FAKE', or 'FALSE', respectively, indicating the classification of the news as described in section "Dataset description and evaluation metrics".

The baseline model's validation and test data results are presented in Table 2, featuring precision, recall, and Macro F1-score. The baseline model incorporates the transformer-based ViT model for images and multilingual BERT (mBERT) for headline and news text. Additionally, we have studied the DeiT and multilingual RoBERTA (XLMRoBERTa) models in our experiments, along with different combinations of

Table 2 Results of baseline models

Data	Image model	Text model	Precision	Recall	F1-Score
Validation	ViT	mBERT	0.8609	0.8555	0.8581
	ViT	XLMRoBERTa	0.8479	0.8552	0.8514
	DeiT	mBERT	0.8678	0.8678	0.8678
	DeiT	XLMRoBERTa	0.8405	0.8429	0.8417
Test	ViT	mBERT	0.8314	0.8413	0.8360
	ViT	XLMRoBERTa	0.8281	0.8526	0.8380
	DeiT	mBERT	0.8592	0.8670	**0.8629**
	DeiT	XLMRoBERTa	0.8371	0.8299	0.8333

The best value among the different models shown in bold

Table 3 Classwise result of best-performing baseline model

Model	Data	Class	Precision	Recall	F1-Score
mBERT+DeIT	Validation	TRUE	1.0000	1.0000	1.0000
		FAKE	0.9183	0.9183	0.9183
		FALSE	0.6852	0.6852	0.6852
		OVERALL	0.8678	0.8678	0.8678
	Test	TRUE	1.0000	1.0000	1.0000
		FAKE	0.9197	0.9065	0.9130
		FALSE	0.6579	0.6944	0.6757
		OVERALL	0.8592	0.8670	0.8629

these models. The key findings from the analysis of the baseline results are summarized as follows.

- We can see that the combination of DeiT+mBERT has given the best overall validation and test results for our dataset with an F1-score of 0.8678 and 0.8629 respectively.
- Moreover, we can see that the validation data the ViT+mBERT model was able to achieve a close score near to DeiT+mBERT model, but it was performing well for test data.
- For the test data, we can see all the different combination of model are behaving more or less same apart from the DeiT+mBERT model with overall F1-score.

Table 3 provides the classwise results of the best-performing models. The results show that the ’FALSE’ class notably impacts the model’s overall performance, primarily due to the limited number of available samples. However, the validation and test data results are similar, indicating that the model performs well with unseen data. Additionally, the model performed well without incorporating any extra meta information, as it effectively captured essential features from headlines, news, and images. Consequently, we decided to further experiment with the best-performing model by incorporating LLM-generated descriptions, which will be discussed in the next section.

Table 4 Results of the model with images supporting the news

Data	Class	Precision	Recall	F1-Score
Validation	TRUE	1.0000	1.0000	1.0000
	FAKE	0.8376	0.9423	0.8869
	FALSE	0.5714	0.2963	0.3902
	Overall	0.8030	0.7462	0.7590
Test	TRUE	1.0000	1.0000	1.0000
	FAKE	0.8544	0.9712	0.9091
	FALSE	0.7647	0.3611	0.4906
	Overall	0.8730	0.7774	0.7999

Table 5 Results of the model with LLM generated description of the images

Data	Class	Precision	Recall	F1-Score
Validation	TRUE	1.0000	1.0000	1.0000
	FAKE	0.9365	0.8510	0.8917
	FALSE	0.5753	0.7778	0.6614
	Overall	0.8373	0.8762	0.8510
Test	TRUE	1.0000	1.0000	1.0000
	FAKE	0.9187	0.8129	0.8626
	FALSE	0.5000	0.7222	0.5909
	Overall	0.8602	0.8451	0.8178

Results of proposed models on LLM generated descriptions

This section discusses the multimodal fusion model, which incorporates the image descriptions generated by LLM as discussed in section "Description of images using LLM". Our main experiments focused on fusing multimodal data by using supporting images, LLM-generated descriptions, and combinations of both images and their descriptions. Table 4 shows the results of the multimodal fusion model, which integrates the supporting images from the news. In this model, images are transformed into features as explained in section "Image feature extraction", fused with the multimodal data consisting of news, headlines, and images. The table displays the validation and test data results categorized by class. Below are the key observations from these results.

- We can see that the overall results are reduced to nearly 10% and 6% to the validation and test data.
- The addition of Image features has not helped model to learn anything new.

We further conducted an experiment involving the description of the images produced by LLMs. Just as we had included the text features of the news headline and content, we also extracted mBERT features for the description. These features were combined with the multimodal features and used for training in the model. Table 5 shows the validation and test data results of the fusion model, which considered the image description text only. The following are key observation of the results.

- We can see that the result has improved compared to the image-only model; this can be because of the better description generated by the LLMs, and the model was able to capture that information better than from the image.
- The result shows that 'FALSE' class scores have increased by almost 50% compared to the image-only model. This has helped in the overall score.

We conducted another experiment by combining the image and text features generated by the LLMs with multimodal features. As mentioned earlier, we acquired the image features from DeiT and the text features from mBERT, which was additionally trained for prediction. The results from this model are summarized in Table 6. Upon analyzing the results, the following observations were made.

- We can see that the results have improved compared to the image-only model but not the text-only one.
- The result of 'FAKE' is increased by 2–3% compared to the other model, whereas the 'FALSE' has caused a decrease in the overall score.
- Also, we can observe that the supporting image and its text generated by LLMs have helped to improve the overall results of the model.

The experiments above show that providing additional information to the model helps improve its performance, especially when we have the textual descriptions. This is likely because the underlying models struggle to understand the content in the images that are given as supporting information for the classes. Moreover, upon reviewing the descriptions generated by LLM, it is evident that the model successfully captured the necessary information for our proposed model to make accurate predictions.

Results of proposed models with representations from Siamese networks

In this section, we discuss the results of the proposed model, which involved incorporating additional features from the Siamese network as described in section Feature extraction using Siamese network. The primary motive behind integrating the network was to leverage the improvements achieved in the model performance by incorporating image descriptions. Moreover, we can see a relation between the headline and the description, which can help the model learn distinct features. We used the same combination

Table 6 Results of the model with both LLM generated description along with the images

Data	Class	Precision	Recall	F1-Score
Validation	TRUE	1.0000	1.0000	1.0000
	FAKE	0.8904	0.9375	0.9133
	FALSE	0.6977	0.5556	0.6186
	Overall	0.8627	0.8310	0.8440
Test	TRUE	1.0000	1.0000	1.0000
	FAKE	0.9155	0.9353	0.9253
	FALSE	0.7273	0.6667	0.6957
	Overall	0.8809	0.8673	0.8736

Table 7 Results of the model with representations from Siamese networks

Data	Class	Precision	Recall	F1-Score
Validation	TRUE	1.0000	1.0000	1.0000
	FAKE	0.9037	0.9471	0.9249
	FALSE	0.7500	0.6111	0.6735
	Overall	0.8846	0.8527	0.8661
Test	TRUE	1.0000	1.0000	1.0000
	FAKE	0.9085	0.9281	0.9181
	FALSE	0.6970	0.6389	0.6667
	Overall	0.8685	0.8556	0.8616

mBERT+DeiT model for this experiment and achieved an F1-score of 0.8616 for the test data. Table 7 shows the classwise results of the proposed method.

Based on the outcomes, it is evident that the representations or vectors from the Siamese network have enhanced the model's learning capabilities, outperforming several other proposed models, particularly those incorporating image descriptions. Notably, the results closely align with the baseline model, which solely leveraged text and image features. Here are some observations derived from the results. There is a significant increase in the classwise result for the model compared with other proposed methods.

Error analysis of proposed models and XAI-based reasoning

In this section, we conducted an analysis of both correctly classified and misclassified samples using our best-proposed models. We selected samples from our test dataset that the model had not seen and then evaluated the results. We focused on two of our top-performing models: mBERT+DeiT (referred to as Prediction (mBERT+DeiT)), which serves as the baseline, and the model that integrated image description and their text (referred to as Predicted(Im+Desc)). The respective results are presented in Tables 8 and 9.

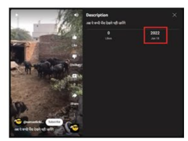
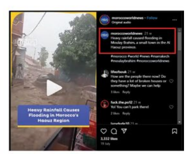
The baseline model makes predictions without the support of any additional information from text or images, leading to both correct and incorrect predictions. For instance, in Table 8, we can see that the baseline model has one prediction as correct, and the LLM-based image description model was able to get two predictions correctly. Moreover, the proposed LLM-based model is able to give a correct prediction for the challenging case of the 'FALSE' class. When coming to the second sample, the model's prediction went wrong, which could be because of the lack of information from images. In the third sample, both the models have incorrect predictions, which can be because of the lack of any support images and information from the text data. Moreover, we can see that for the second and fourth samples, the model with image descriptions has helped to give correct predictions.

The key observation from the analysis is that the baseline model with only text and image was able to give correct predictions when the context was relevant. But when there is additional information is required to support the data, it fails and gives false positives. Similarly, in the case of the LLM-based model, the prediction goes wrong when the image description doesn't contain any relevant information connecting to the context of the data, as seen in samples 1 and 3 in Table 8.

Table 8 Results of our best proposed models (mBERT+DeiT & Img+Desc) sample set 1

Headline	News	Image	Support-Image	Original Label	Prediction (mBERT+DeiT)	Prediction (Im+Desc)
விஜய் படம் முதல் சன் டிவி சீரியல் வரை நடித்த இவர் தான் பிரபல நடிகர் நம்பியாரின் மகள்! இவர் தமிழ் மற்றும் மலையாள மொழி படங்களில் நடித்திருக்கிறார். தமிழில் தளபதி விஜய் நடிப்பில் வெளிவந்த ஆதி படத்திலும், விஷாலின் வெடிப்படத்திலும் குணச்சித்திர வேடத்தில் நடித்திருக்கிறார்.	பிரபல நடிகர் நம்பியாரின் மகள் தான் சீரியல் நடிகை சினேகா என்று கூறி புகைப்படம் .. மேலும் அப்பதிவுகளில், "... எம்.என்.நம்பியாரின் மகள் என்று சமூக ஊடகங்களில் பரவும் செய்திகள் தவறானவை என்பதை அறிய முடிகிறது.			FALSE	FALSE	FAKE
முதல்வர் ஸ்டாலின் அவர்களே சொன்னா கோச்சிக்கிறீங்க இருந்தாலும் சொல்லித்தான் ஆகணும். உங்க அரசாங்கம் உண்மையிலேயே கையாளாகாத அரசாங்கம் தான்.	இந்த பள்ளி மாணவிகளுக்கு எந்த பள்ளியில் படிக்கின்றோம் என்பதே தெரியவில்லை, ...			FALSE	FAKE	FALSE
பாவம் கார்கே.. அவர் பேசும்போது மோடி மோடி என்று கூவி அவரை பதற்றமடைய செய்கிறார்கள்..	இந்திய தேசிய காங்கிரஸின் தலைவரான மல்லிகார்ஜுன கார்கே மேடை ஒன்றில் பேசும் போது, ..., கார்கே பேசும்போது மோடி மோடி என்று மக்கள் கூச்சலிடுவதாகப்.		-	FAKE	FALSE	FALSE
அன்று இந்த அண்ணாமலை இப்படி வளர்ந்து வருவார் என்று சன் டிவியும் சரி, திமுக கொத்தடிமைகளும் சரி கனவில் கூட நினைத்திருக்க மாட்டார்கள்!!	பாரதிய ஜனதா கட்சியின் மாநில தலைவர் அண்ணாமலை சூப்பர் ஸ்டார் ரஜினி காந்தின் படையப்பா படம் குறித்து ரஜினி முன்னிலையில் உரையாற்றியதாக இளைஞர் ஒருவர் பேசக்கூடிய பழைய வீடியோ ஒன்று சமூக வலைத்தளங்களில் பாஜக ஆதரவாளர்களால் பரப்பப்பட்டு வருகிறது...			FAKE	FALSE	FAKE

Table 9 Results of our best proposed models (mBERT+DeiT & Img+Desc) sample set 2

Headline	News	Image	Support-Image	Original Label	Prediction (mBERT+DeiT)	Prediction (Im+Desc)
பெங்களூருவில் இருந்து சென்னைக்கும்,சென்னையில் இருந்து பெங்களூருவுக்கும் பயணிக்கும் அரசுப் பேருந்து! திராவிட மாடல் எப்படி?	பாஜக மாநில செயற்குழு உறுப்பினர் சௌதா மணி மற்றும் .. தமிழ்நாடு அரசு பேருந்தின் நிலை மோசமாக இருப்பதாக பெங்களூர் மற்றும் தமிழ்நாடு அரசு பேருந்துகளின் புகைப்படங்களை ஒப்பிட்டு டிவிட் செய்துள்ளார்.			FAKE	FALSE	FAKE
காங்கிரஸ் கொடியை கண்டாலே எருமை மாடுகளுக்கு கூட பிடிக்கல, மனிதர்களுக்கு எப்படி பிடிக்கும்	கையில் காங்கிரஸ் கட்சிக் கொடியுடன் சென்றவர்களை எருமை மாடு துரத்தி முட்டியதாக வீடியோ ஒன்று சமூக வலைத்தளங்களில் பரப்பப்படுகிறது...			FAKE	FALSE	FAKE
இன்று மேட்ச்... My Man குறிப்பிட்டு சுப்மன் கில்லுக்கு சாரா டெண்டுல்கர் வாழ்த்து..	இந்தியாவில் நடைபெற்று வரும் 13-வது உலகக்கோப்பை கிரிக்கெட் தொடரில், இந்தியா மற்றும் நியூசிலாந்து அணிகளுக்கு இடையிலான முதல் அரையிறுதி போட்டி இன்று மும்பை வான்கிடே மைதானத்தில் நடைபெறவுள்ளது....			FALSE	FAKE	FALSE
மழையின் கோர பிடியில் நெல்லை.	குமரிக்கடல் மற்றும் அதனை ஒட்டியுள்ள பகுதிகளில் வளிமண்டல கீழ் அடுக்கு சுழற்சி நிலவுவதன் காரணமாகத் தூத்துக்குடி, திருநெல்வேலி, கன்னியாகுமரி தென்காசி ஆகிய மாவட்டங்களில்...			FALSE	FAKE	FAKE

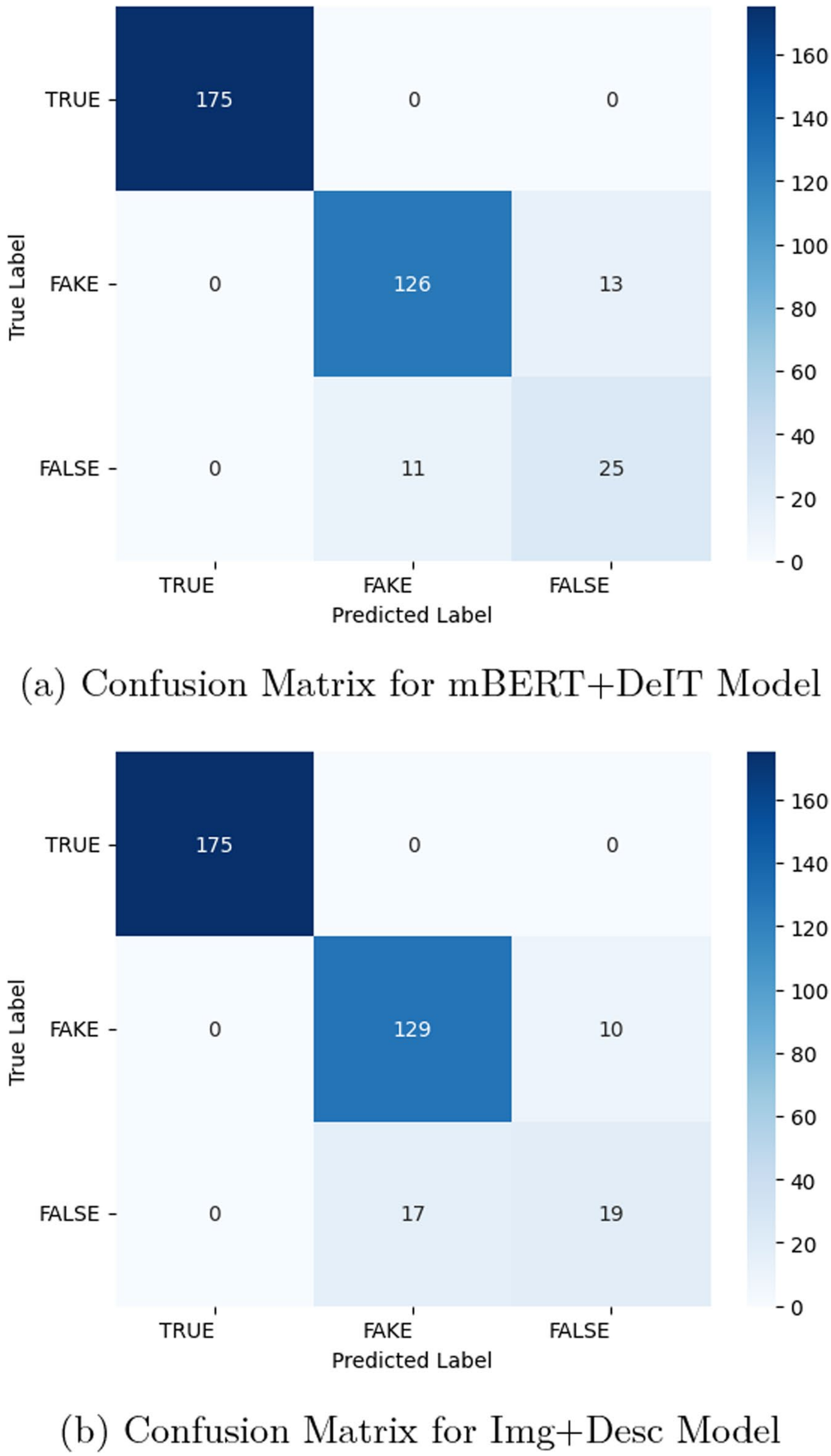
We have additional examples where our proposed model, along with supporting images and their LLM-based description, helped make accurate predictions. Now we consider the samples in Table 9, where we have another set of samples where we analyzed the model predictions. Here, we can see that the image description model has clearly helped in giving the correct predictions with additional features the model has learned. For the first three samples, the model with image description gave a correct prediction, whereas the baseline model with only the image and news data couldn't capture all the details. However, for the last sample, both models gave wrong predictions. The LLM-based model couldn't capture the necessary information from the supported image, which made the model predict wrongly, whereas the baseline model, without additional information, fo the prediction wrong. We have given the confusion matrix of our best proposed models in Fig. 5. We can see from the confusion matrix that the confused classes between 'FALSE' and 'FAKE' are improved for the image description based model.

Based on our analysis, the model's performance will improve with adequate meta information to support or reject the claim, which will assist in labeling the samples. Additionally, the model demonstrated the ability to correctly predict samples by extracting descriptions from supporting images. This is because the text provided much of the information that could not be gathered from the image.

Explainable AI based reasoning on proposed methods

In this section, we discuss the proposed model's capability to understand and interpret the data. We use the XAI methodology to analyze the results of the proposed model. LIME (Local Interpretable Model-agnostic Explanations) [40] is a method that aims to explain the predictions of any machine learning model by approximating the model's behavior locally around a specific instance. The main idea is to create an interpretable model, like a linear model, that can imitate the complex model's behavior near the instance being explained. This process starts by choosing the instance for which an explanation is needed and then generating a dataset of perturbed samples around this instance. The complex model is used to predict outcomes for these perturbed samples. The coefficients or structure of the interpretable model provide insights into how the complex model predicted the original instance. For example, if the interpretable model is a linear regression, the coefficients indicate the contribution of each feature to the prediction. This approach ensures that the explanation accurately represents the model's behavior in the specific region around the instance. LIME's flexibility allows it to be used with various models and data, making it a powerful tool for improving the transparency and interpretability of machine learning applications. However, the quality of the explanation depends on how well the interpretable model approximates the complex model locally, and creating perturbations can be computationally expensive.

Since our model is multimodal, the interpretations by LIME are possible with image and text but not as multimodal. Hence, we get the explanations for the text, match it with the corresponding image, and show how the predictions are relevant. Moreover, we have used the images that support the claims for interpretation and the LLM-based descriptions. We have performed LIME interpretation for the text-based



(a) Confusion Matrix for mBERT+DeIT Model

(b) Confusion Matrix for Img+Desc Model

Fig. 5 Confusion matrix of best proposed models

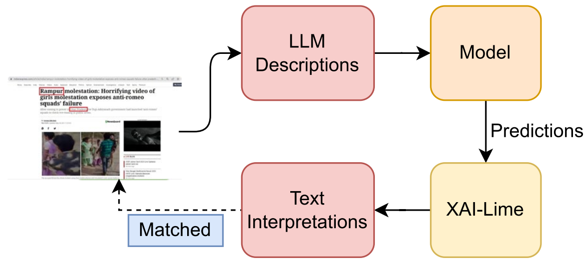
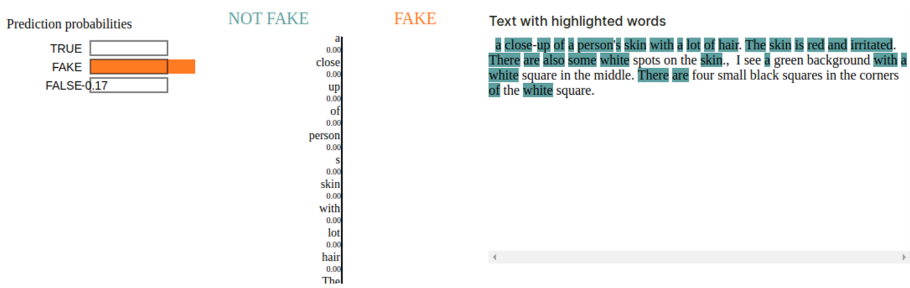


Fig. 6 LIME for image matching



(a) Lime Prediction

(b) Support Image

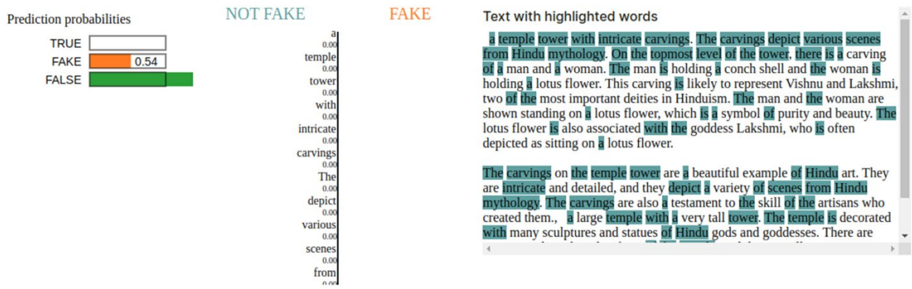
Fig. 7 LIME interpretations: sample I

description and have identified the corresponding content in the image that supports the claim.

The samples given in the figures above show the LIME-generated explanations for some of the samples. We can see that the LLM-generated descriptions have helped the model to learn the context, which has caused the claim to be fake or false (Figs. 6, 7, 8, 9, 10, 11).

Conclusions and future works

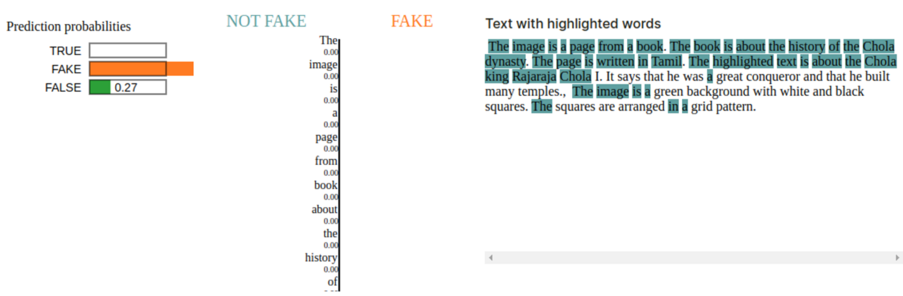
In this work, we have proposed a multimodal fake news detection dataset for the Tamil language and a baseline fusion model that considers the news content, headlines, and images. Moreover, we have introduced the LLM capability to address fake news detection by incorporating the description generated by LLM as an additional feature to train our proposed baseline model. To this end, we proposed a multimodal fake news dataset for Tamil and a baseline model incorporating additional information from the data.



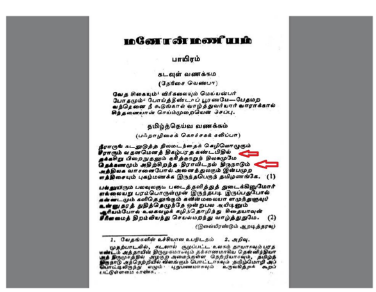
(a) Lime Prediction

(b) Support Image

Fig. 8 LIME interpretations: sample II



(a) Lime Prediction



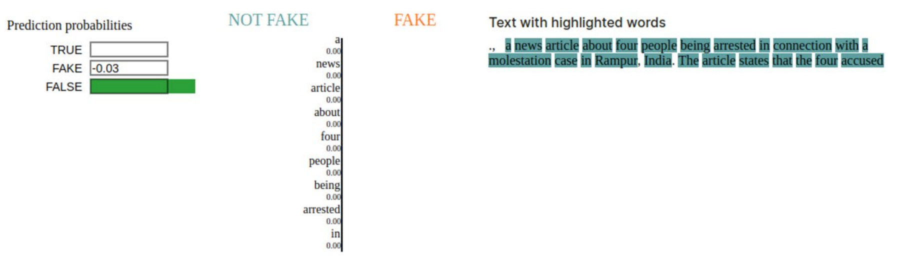
(b) Support Image

Fig. 9 LIME interpretations: sample III

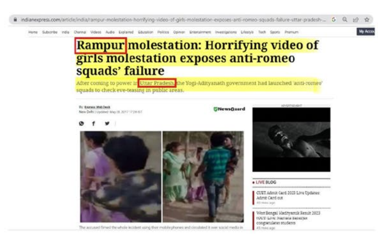
(a) Lime Prediction

(b) Support Image

Fig. 10 LIME interpretations: sample IV



(a) Lime Prediction



(b) Support Image

Fig. 11 LIME interpretations: sample V

Our proposed multimodal fusion model incorporating the news headlines, content, and images has achieved an overall F1-score of 0.8629, and the model incorporating LLM-based description has achieved an overall F1-score of 0.8736. Moreover, we added the additional features using Siamese networks, which resulted in an F1-score of 0.8616.

In our study, we found that combining multimodal content, meta information, and LLM-generated description led to performance improvement. However, it had an impact on a class with fewer samples, as some samples belonged to the 'FALSE' class and contained images that were challenging for the model to interpret. While the description provided some information to the model, it could not effectively differentiate between the labels. In the future, we plan to investigate further the various factors influencing the authenticity of news content and how supporting information can impact news propagation.

Currently, data on social media is being presented in various formats, including video with text and audio with text. We aim to expand our research into these different modalities. The main challenge will be collecting the necessary data to advance our research. While there are limited resources that provide open access to data, we can utilize fact-checking websites to obtain verified news articles. However, news in different formats is not readily available, as verifying this content requires more extensive analysis and evidence. Traditional methods like data augmentation can help address data imbalance issues. However, the challenge we face is working with real-time data retrieved from social media. Using augmented data may limit the model's exposure to the same information instead of allowing it to learn from diverse scenarios. Additionally, synthetic data may not be adequate when testing against real-world data.

Author contributions
Hariharan RamakrishnaIyer LekshmiAmmal—Conceptualization, Methodology, Data curation, software, and Writing - Original Draft. Anand Kumar Madasamy—Conceptualization, Methodology, Writing—Review & Editing, Supervision.

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Data availability
The data used for the current study will be made available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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